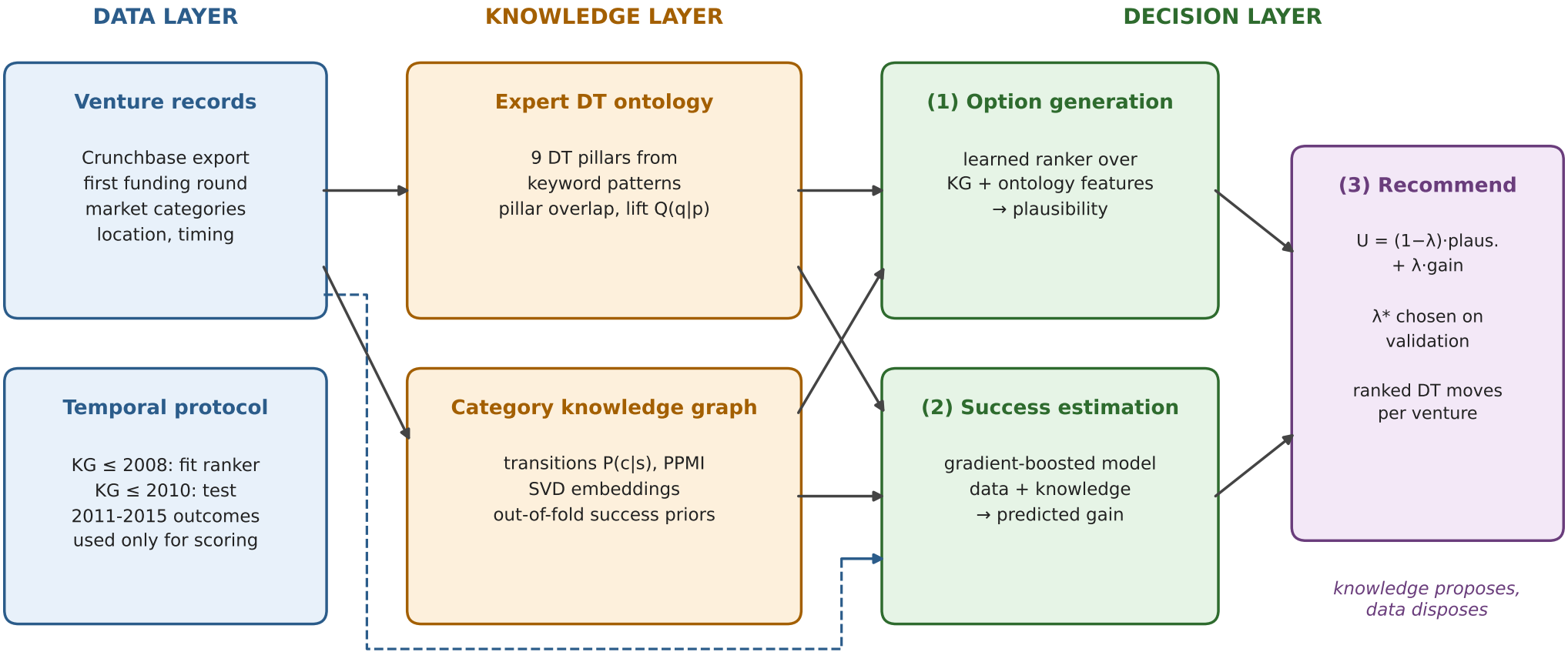
DATA LAYER
KNOWLEDGE LAYER
DECISION LAYER
Venture records
Crunchbase export
first funding round
market categories
location, timing
Temporal protocol
KG ≤ 2008: fit ranker
KG ≤ 2010: test
2011-2015 outcomes
used only for scoring
Expert DT ontology
9 DT pillars from
keyword patterns
pillar overlap, lift Q(q|p)
Category knowledge graph
transitions P(c|s), PPMI
SVD embeddings
out-of-fold success priors
(1) Option generation
learned ranker over
KG + ontology features
→ plausibility
(2) Success estimation
gradient-boosted model
data + knowledge
→ predicted gain
(3) Recommend
U = (1−λ)·plaus.
+ λ·gain
λ* chosen on
validation
ranked DT moves
per venture
knowledge proposes,
data disposes